

Cheap Public Transport API (0.1.0)

Download OpenAPI specification:

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This API find the cheapest journey between UK cities via public transport. This combines national rail services, FlixBus coaches, and National Express coaches.

Travel Routes

The individual intercity legs which make up a Journey.

Get All Routes

List all travel routes

Retrieves every available route between stations from the database.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of all Route objects.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

Notes:

- **Caution:** This dataset contains ~20,000 records. Testing this in a browser may cause performance issues or crashes.

AUTHORIZATIONS: > *HTTPBearer*

Responses

> 200 Successful Response

GET /routes/

Response samples

200

Content type

application/json

Copy Expand all Collapse all

```
[
  - {
    "price": 12.5,
    "notes": "Northern Rail Service",
    "id": 1,
    + "origin_station": { ... },
    + "destination_station": { ... },
    + "transport_mode": { ... }
  }
]
```

Create New Route

Create a new travel route

Adds a new route record after validating that the stations and transport mode exist.

Access Level: - Admin only.

Args:

- **route:** Schema containing origin/destination station IDs, transport mode ID, and price.

- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User does not have administrative privileges.
- **404**: Not Found - If the origin station, destination station, or transport mode does not exist.

AUTHORIZATIONS: > HTTPBearer

REQUEST BODY SCHEMA: application/json

required

price > required	Price (number) or Price (string) (Price)
notes > required	Notes (string) or Notes (null) (Notes)
origin_station_id required	integer (Origin Station Id)
destination_station_id required	integer (Destination Station Id)
transport_mode_id required	integer (Transport Mode Id)

Responses

> 201 Successful Response

> 422 Validation Error

POST /routes/

Request samples

Payload

Content type
application/json

Copy

```
{
  "price": 12.5,
  "notes": "Northern Rail Service",
  "origin_station_id": 0,
  "destination_station_id": 0,
  "transport_mode_id": 0
}
```

Response samples

201

422

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "price": 12.5,
  "notes": "Northern Rail Service",
  "id": 1,
  - "origin_station": {
    "name": "Leeds Station",
    "city_id": 1,
    + "city": { ... }
  },
  - "destination_station": {
    "name": "Leeds Station",
    "city_id": 1,
    + "city": { ... }
  },
  - "transport_mode": {
    "name": "train",
    "id": 1
  }
}
```

Get Route By Id

Get specific route details

Fetches the full details of a single travel route using its unique ID.

Access Level: - Admin and User.

Args:

- **route_id:** The unique identifier of the route.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - No route exists with the provided ID.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

route_id	integer (Route Id)
required	

Responses

> **200** Successful Response

— **404** Route not found

> **422** Validation Error

GET /routes/{route_id}

Response samples

200

422

Content type

application/json

Copy Expand all Collapse all

```
{
  "price": 12.5,
  "notes": "Northern Rail Service",
  "id": 1,
```

```
- "origin_station": {
  "name": "Leeds Station",
  "city_id": 1,
  + "city": { ... }
},
- "destination_station": {
  "name": "Leeds Station",
  "city_id": 1,
  + "city": { ... }
},
- "transport_mode": {
  "name": "train",
  "id": 1
}
}
```

Delete Route

Delete a route

Permanently removes a specific route record from the database.

Access Level: - Admin only.

Args:

- **route_id:** The unique identifier of the route to be deleted.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.
- **404:** Not Found - No route exists with the provided ID.

Notes:

- This action is destructive but does not affect associated station or city records.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

route_id	integer (Route Id)
required	

Responses

— **204** Successful Response

— **404** Route not found

> **422** Validation Error

DELETE /routes/{route_id}

Response samples

422

Content type

application/json

Copy Expand all Collapse all

```
{
  - "detail": [
    + { ... }
  ]
}
```

Update Route

Update an entire route

Replaces an existing route's data. Validates the existence of related entities before saving.

Access Level: - Admin only.

Args:

- **route_id:** The unique identifier of the route to update.
- **route_update:** The new data to apply to the route.

- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User does not have administrative privileges.
- **404**: Not Found - If the route, either station, or the transport mode is not found.

AUTHORIZATIONS: > HTTPBearer

PATH PARAMETERS

route_id required	integer (Route Id)
----------------------	--------------------

REQUEST BODY SCHEMA: application/json
required

price > required	Price (number) or Price (string) (Price)
---------------------	--

notes > required	Notes (string) or Notes (null) (Notes)
---------------------	--

origin_station_id required	integer (Origin Station Id)
-------------------------------	-----------------------------

destination_station_id required	integer (Destination Station Id)
------------------------------------	----------------------------------

transport_mode_id required	integer (Transport Mode Id)
-------------------------------	-----------------------------

Responses

> 200 Successful Response

— 404 Route not found

> 422 Validation Error

PUT /routes/{route_id}

Request samples

Payload**Content type**

application/json

[Copy](#)

```
{
  "price": 12.5,
  "notes": "Northern Rail Service",
  "origin_station_id": 0,
  "destination_station_id": 0,
  "transport_mode_id": 0
}
```

Response samples**200****422****Content type**

application/json

[Copy](#)[Expand all](#)[Collapse all](#)

```
{
  "price": 12.5,
  "notes": "Northern Rail Service",
  "id": 1,
  - "origin_station": {
    "name": "Leeds Station",
    "city_id": 1,
    + "city": { ... }
  },
  - "destination_station": {
    "name": "Leeds Station",
    "city_id": 1,
    + "city": { ... }
  },
  - "transport_mode": {
    "name": "train",
    "id": 1
  }
}
```

The 200 largest UK cities with latitude and longitude.

Get All Cities

Get all cities

Retrieves a list of every city currently stored in the database.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of City objects.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

AUTHORIZATIONS: > *HTTPBearer*

Responses

> **200** Successful Response

GET /city/

Response samples

200

Content type

application/json

Copy

Expand all

Collapse all

```
[
  - {
    "name": "Leeds",
    "latitude": 53.79,
    "longitude": 1.61,
    "id": 1
  }
]
```

Create New City

Create a new city

Adds a unique city record to the database.

Access Level: - Admin only.

Args:

- **city**: Schema containing name, latitude, and longitude.
- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User does not have administrative privileges.
- **422**: Validation Error - Input data is invalid.

Notes:

- City names are automatically converted to lowercase.
- The name field must be unique.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

required

name

required

string (Name)

latitude > required	Latitude (number) or Latitude (string) (Latitude)
longitude > required	Longitude (number) or Longitude (string) (Longitude)

Responses

> **201** Successful Response

> **422** Validation Error

POST /city/

Request samples

Payload

Content type
application/json

Copy

```
{
  "name": "Leeds",
  "latitude": 53.79,
  "longitude": 1.61
}
```

Response samples

201

422

Content type
application/json

Copy

```
{
  "name": "Leeds",
  "latitude": 53.79,
```

```
    "longitude": 1.61,  
    "id": 1  
  }
```

Get City By Id

Get city by ID

Fetches a specific city's details using its unique integer ID.

Access Level: - Admin and User.

Args:

- **city_id**: The unique identifier of the city.
- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User role not recognized.
- **404**: Not Found - No city exists with the provided ID.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

city_id required	integer (City Id)
---------------------	-------------------

Responses

> **200** Successful Response

— **404** City not found

> **422** Validation Error

GET /city/id/{city_id}

Response samples

200

422

Content type

application/json

Copy

```
{
  "name": "Leeds",
  "latitude": 53.79,
  "longitude": 1.61,
  "id": 1
}
```

Get City By Name

Get city by name

Fetches a specific city's details using its name.

Access Level: - Admin and User.

Args:

- **city_name:** The name string of the city.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - No city exists with the provided name.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

city_name
required

string (City Name)

Responses

> **200** Successful Response

— **404** City not found

> **422** Validation Error

GET /city/name/{city_name}

Response samples

200

422

Content type

application/json

Copy

```
{
  "name": "Leeds",
  "latitude": 53.79,
  "longitude": 1.61,
  "id": 1
}
```

Update City

Update city

Modifies an existing city record based on its ID.

Access Level: - Admin only.

Args:

- **city_id:** The unique identifier of the city to update.

- **city_update**: The new data to apply to the city.
- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User does not have administrative privileges.
- **404**: Not Found - The city to update is not found.

AUTHORIZATIONS: > HTTPBearer

PATH PARAMETERS

city_id required	integer (City Id)
---------------------	-------------------

REQUEST BODY SCHEMA: application/json
required

name required	string (Name)
------------------	---------------

latitude > required	Latitude (number) or Latitude (string) (Latitude)
------------------------	---

longitude > required	Longitude (number) or Longitude (string) (Longitude)
-------------------------	--

Responses

> 200 Successful Response

— 404 City not found

> 422 Validation Error

PUT /city/{city_id}

Request samples

Payload

Content type

application/json

[Copy](#)

```
{
  "name": "Leeds",
  "latitude": 53.79,
  "longitude": 1.61
}
```

Response samples

200

422

Content type

application/json

[Copy](#)

```
{
  "name": "Leeds",
  "latitude": 53.79,
  "longitude": 1.61,
  "id": 1
}
```

Stations

Get All Stations

Get all stations

Retrieves a complete list of all stations stored in the database.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of Station objects.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

AUTHORIZATIONS: > *HTTPBearer*

Responses

> **200** Successful Response

GET /stations/

Response samples

200

Content type

application/json

Copy Expand all Collapse all

```
[
  - {
      "name": "Leeds Station",
      "city_id": 1,
    + "city": { ... }
  }
]
```

Create New Station

Create a new station

Adds a new station record to the database after verifying the associated city exists.

Access Level: - Admin only.

Args:

- **station:** Schema containing the station name and city ID.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.
- **404:** Not Found - If the provided `city_id` does not match any existing city.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

required

name required	string (Name)
------------------	---------------

city_id required	integer (City Id)
---------------------	-------------------

Responses

> **201** Successful Response

— **404** City not found

> **422** Validation Error

POST /stations/

Request samples

Payload

Content type

application/json

[Copy](#)

```
{
  "name": "Leeds Station",
  "city_id": 1
}
```

Response samples

201

422

Content type

application/json

[Copy](#)[Expand all](#)[Collapse all](#)

```
{
  "name": "Leeds Station",
  "city_id": 1,
  - "city": {
    "name": "Leeds",
    "latitude": 53.79,
    "longitude": 1.61,
    "id": 1
  }
}
```

Get Station By Id

Get station by ID

Fetches the details of a specific station using its unique integer ID.

Access Level: - Admin and User.

Args:

- **station_id:** The unique identifier of the station.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - If no station exists with the provided ID.

AUTHORIZATIONS: > HTTPBearer

PATH PARAMETERS

station_id required	integer (Station Id)
------------------------	----------------------

Responses

> 200 Successful Response

— 404 Station not found

> 422 Validation Error

GET /stations/{station_id}

Response samples

200

422

Content type
application/json

Copy Expand all Collapse all

```
{
  "name": "Leeds Station",
  "city_id": 1,
  - "city": {
    "name": "Leeds",
    "latitude": 53.79,
    "longitude": 1.61,
    "id": 1
  }
}
```

Categorises routes based on transport mode such as 'train' or 'coach'.

Get All Transport Mode

Retrieve all transport modes

Fetches a complete list of all available transport categories.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of TransportMode objects, useful for UI filters or dropdown menus.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

AUTHORIZATIONS: > *HTTPBearer*

Responses

> **200** Successful Response

GET /transport_mode/

Response samples

200

Content type

application/json

[Copy](#) [Expand all](#) [Collapse all](#)

```
[
  - {
    "name": "train",
    "id": 1
  }
]
```

Create New Transport Mode

Register a new transport mode

Adds a new category of transport (e.g., Train, Coach) to the system.

Access Level: - Admin only.

Args:

- **transport_mode:** Schema containing the name of the transport mode.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.

Notes:

- This creates a parent category that individual routes must be associated with.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

required

name	string (Name)
required	

Responses

> **201** Successful Response

> **422** Validation Error

POST /transport_mode/

Request samples

Payload

Content type

application/json

Copy

```
{
  "name": "train"
}
```

Response samples

201

422

Content type

application/json

Copy

```
{
  "name": "train",
  "id": 1
}
```

Get Transport Mode By Id

Get transport mode by ID

Retrieves the details of a specific transport category using its unique ID.

Access Level: - Admin and User.

Args:

- **transport_mode_id:** The unique identifier of the transport mode.

- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User role not recognized.
- **404**: Not Found - If no transport mode exists with the provided ID.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

transport_mode_id	integer (Transport Mode Id)
<i>required</i>	

Responses

> **200** Successful Response

> **422** Validation Error

GET /transport_mode/{transport_mode_id}

Response samples

200

422

Content type

application/json

Copy

```
{
  "name": "train",
  "id": 1
}
```

Journeys

A collection of Travel Routes which makes a journey between any two given cities. This will return the cheapest possible combination of Travel Routes to complete a journey, as well as cost breakdown.

Get Journey

Fetch a journey by city IDs

Calculates the cheapest route between two cities using their unique integer IDs.

Access Level: - Admin and User.

Args:

- **origin_id:** ID of the starting city.
- **destination_id:** ID of the destination city.
- **railcard_discount:** Percentage discount to apply (default 0).
- **advanced_fares:** Whether to include advanced fare pricing estimate.
- **max_coach_legs:** Maximum number of coach segments allowed.

Returns:

- A breakdown of total costs and a list of journey legs.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - No journey could be calculated between the provided IDs.

Notes:

- Total price includes a £2 changeover fee for station transfers.
- Returns the full path including all journey legs.

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

origin_id required	integer (Origin Id)
destination_id required	integer (Destination Id)
railcard_discount	integer (Railcard Discount) Default: 0

advanced_fares	boolean (Advanced Fares) Default: false
max_coach_legs	integer (Max Coach Legs) Default: 2

Responses

> **200** Successful Response

> **422** Validation Error

GET /journeys/id

Response samples

200

422

Content type
application/json

Copy Expand all Collapse all

```
{
  "total_price": 12.5,
  "ticket_price": 12.5,
  "railcard_discounts": 0,
  "advanced_discounts": 0,
  "transfer_price": 0,
  - "path": [
    + { ... }
  ]
}
```

Get Journey

Fetch a journey by city names

Calculates the cheapest route between two cities using their string names.

Access Level: - Admin and User.

Args:

- **origin_name:** Name of the starting city (case-insensitive).
- **destination_name:** Name of the destination city (case-insensitive).
- **railcard_discount:** Percentage discount to apply (default 0).
- **advanced_fares:** Whether to include advanced fare pricing estimate.
- **max_coach_legs:** Maximum number of coach segments allowed.

Returns:

- A breakdown of total costs and a list of journey legs.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - If either city name does not exist or no path is found.

Notes:

- Total price includes a £2 changeover fee for station transfers.

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

origin_name <i>required</i>	string (Origin Name)
destination_name <i>required</i>	string (Destination Name)
railcard_discount	integer (Railcard Discount) Default: 0
advanced_fares	boolean (Advanced Fares) Default: false
max_coach_legs	integer (Max Coach Legs) Default: 2

Responses

> **200** Successful Response

> **422** Validation Error

GET /journeys/name

Response samples

200

422

Content type

application/json

Copy Expand all Collapse all

```
{
  "total_price": 12.5,
  "ticket_price": 12.5,
  "railcard_discounts": 0,
  "advanced_discounts": 0,
  "transfer_price": 0,
  - "path": [
    + { ... }
  ]
}
```

Authentication

Endpoints to authenticate users via Supabase.

Signup

Register a new account

Creates a new user record in Supabase with a default 'user' role.

Args:

- **data**: Schema containing the user's email and password.

Returns:

- A success message and the unique Supabase user ID.

Errors:

- **400**: If the email is already registered or the password does not meet requirements.

REQUEST BODY SCHEMA: application/json

required

email required	string <email> (Email)
-------------------	------------------------

password required	string (Password)
----------------------	-------------------

Responses

> **200** Successful Response

> **422** Validation Error

POST /auth/signup

Request samples

Payload

Content type
application/json

Copy

```
{
  "email": "user@example.com",
  "password": "string"
}
```

Response samples

200

422

Content type

application/json

Copy

```
{
  "message": "string",
  "id": "string"
}
```

Login

Authenticate user

Verifies credentials and returns a set of JWT tokens for subsequent requests.

Args:

- **data**: Schema containing the user's email and password.

Returns:

- **access_token**: The Bearer token used for authorization headers.
- **refresh_token**: Used to obtain a new access token when it expires.
- **token_type**: Always returns 'bearer'.

Errors:

- **401**: If the email or password credentials are incorrect.

REQUEST BODY SCHEMA: application/json

required

email required	string <email> (Email)
-------------------	------------------------

password required	string (Password)
----------------------	-------------------

Responses

> **200** Successful Response

> **422** Validation Error

POST /auth/login

Request samples

Payload

Content type

application/json

Copy

```
{
  "email": "user@example.com",
  "password": "string"
}
```

Response samples

200

422

Content type

application/json

Copy

```
{
  "access_token": "string",
  "refresh_token": "string",
  "token_type": "bearer"
}
```